Research Plan

Arjun Paradkar, Reva Paradkar, Arham Shah

Topology-Aware Physics-constrained Operator Learning of Eigenmode Instability Spectra in Vlasov-Maxwell Plasma Systems for Anomalous Energy Transport Suppression in Fusion Plasmas.

Accuracy and Methods Under Review by University of Arkansas Physics Faculty

Rationale:

Small microscopic instabilities in magnetically confined fusion plasmas (in magnetic confinement fusion reactor prototypes), show up as eigenmods (similar to eigenvalues) of the linearized maxwell vlasov equations. These instabilities are precursors to turbulence (when the plasma goes out of control), and better transport (movement of plasma). Predicting the instability spectrum (made of the various eigenmodes) across the incredible amounts of initial plasma conditions is expensive computationally as it requires repeated running of eigenvalue solvers (current methods). Predicting the movement and stability of the entire plasma thus requires significant computational cost. Recent developments in neural operators (specifically fourier neural operators), can be applied to models for partial differential equations (such as our vlasov maxwell system), and can predict pointwise error in the plasma. Nevertheless, these algorithms do not preserve structural features in the instability spectrum, (mostly on the boundaries of differently behaving sectors of plasma) - regime transitions. The incorrect

combination of branches or misclassification occurs regardless of numerical loss. Our project proposes a topology-aware neural operator replacement that preserves the structure of the instability spectra while being magnitudes faster than eigenvalue solvers. This will enable quicker, but still accurate modeling of plasmas to accelerate development of viable fusion energy.

Research Question:

Can a topology lens on a stack of neural operators predict eigenmode instability spectra with near-numerical solver accuracy and preserve the structural regime transitions while being magnitudes faster.

Hypothesis: A Fourier Neural Operator trained with pointwise loss on the spectra will be good at approximating outputs of eigenvalue solvers, but may fail to preserve correctness near the boundaries between regimes(areas of similarly behaving plasma). Incorporating a persistent-homology based layer of constraints will reduce this error and improve the classification of regimes while being accurate and fast.

Materials:

1. Linear Eigenvalue solvers for reference
2. Python
3. FNOs
4. MLP(Multilayer perceptrons) for baseline
5. CNN baseline

6. Enhanced FNOs
7. Persistent homology library (Ripser or Gudhi)
8. Wasserstein distance computation tools
9. RTX 5070 GPU
10. Ryzen 9 CPU
11. Matplotlib/ Seaborn for data visualization

Procedure:

- Construct Dataset (define equilibrium, generate instability spectra using solver, convert eigenvalues into 2D density grids, define training, IID and Boundary sets)
- Train Baseline models(MLP, CNN, FN, FNO variant, evaluate on MSE, MAE, and regime classification)
- Analysis of Structural failure (See where branches are misclassified)
- Topological Integration (Compute persistence diagrams, Make TSC metric using Wasserstein distance between persistence diagrams. Compare to baseline)
- Train the FNO with topology lens(TSC regularization term, retrain, Evaluate)
- Create an OOD test set and find acceleration magnitude.

Data and Statistical Analysis

- MSE (Mean-Squared-Error)
- MAE (Mean Absolute Error)

- Regime classification metric
- TSC(Topological Spectral Consistency)
- Wasserstein distance
- Boundary performance
- OOD curves
- Runtime acceleration magnitude

Figures:

- Sample spectral density
- Real vs MLP vs FNO
- Training curves
- Structural failure example
- Persistence diagram chars
- Boundary robustness plot
- OOD plot
- Speed of computation plot

Risk Assesment:

- There are no human risks/ risks towards the researchers in development
- In the development we might run into these risks (non-human): overfitting, insufficient data, and lack of computational power. Mitigation: Clear Dataset and Splits, Controlled Hyperparameter Tuning, Cross Validation.

Bibliography

(Note: All papers were recommended or found online and read, but not all were used to develop project plan; Main 5 bolded):

Brizard, A. J., & Hahm, T. S. (2007). Foundations of nonlinear gyrokinetic theory. Reviews of Modern Physics, 79(2), 421–468.

Görler, T., Lapillonne, X., Brunner, S., Dannert, T., Jenko, F., Merz, F., & Told, D. (2011). The global version of the gyrokinetic turbulence code GENE. Journal of Computational Physics, 230(18), 7053–7071.

Belli, E. A., Candy, J., & Waltz, R. E. (2010). Fully electromagnetic gyrokinetic eigenmode analysis of tokamak microinstabilities. Physics of Plasmas, 17(11), 112314.

Fransson, J., Citrin, J., & Bourdelle, C. (2023). A fast neural network surrogate model for the eigenvalues of the gyrokinetic linear stability problem based on QuaLiKiz. Physics of Plasmas, 30(12), 123904.

Gillgren, J., Citrin, J., & Bourdelle, C. (2025). Interpretable neural network surrogate models for gyrokinetic growth rates. Physics of Plasmas, 32(5), 052306.

Li, Z., Kovachki, N., Azizzadenesheli, K., Liu, B., Stuart, A., Bhattacharya, K., & Anandkumar, A. (2021). Fourier neural operator for parametric partial differential equations. International Conference on Learning Representations (ICLR).

Cohen-Steiner, D., Edelsbrunner, H., & Harer, J. (2007). Stability of persistence diagrams. Discrete & Computational Geometry, 37(1), 103–120.

Brüel-Gabrielsson, R., Ganapathi-Subramanian, V., Skraba, P., & Guibas, L. (2020). A topology layer for machine learning. Proceedings of the 37th International Conference on Machine Learning (ICML).

Hofer, C., Kwitt, R., Niethammer, M., & Uhl, A. (2017). Deep learning with topological signatures. Advances in Neural Information Processing Systems (NeurIPS), 30.

Chen, C., & Wang, Y. (2021). An efficient algorithm for computing the 1-Wasserstein distance between persistence diagrams. SIAM Journal on Scientific Computing, 43(6), A3701–A3724.

Leykam, D., Andreanov, A., & Flach, S. (2023). Topological data analysis and machine learning for physical systems. Journal of Physics: Complexity, 4(3), 032001.